

RESEARCH PAPER

WILEY

Global Ecology
and BiogeographyA Journal of
Macroecology

Multiscale patterns of rarity in fungi, inferred from fruiting records

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Editor: Michael Borregaard

Abstract

Aim: It is unknown whether fungi show similar trends to other organisms in their macroecological patterns of abundance and spatial distribution. Here, we investigated fungal abundance–occupancy relationships to determine whether fungi that are common at a local scale tend to be more widely distributed.

Location: UK and Switzerland.

Time period: 1950–2014.

Major taxa studied: Fungi.

Methods: We used a local data set of fruiting records of 2,319 species in the UK, accumulated over 65 years, and one from Switzerland of 319 species, spanning 32 years. Using the number of records and occurrence as proxies for abundance, in each case we examined the form of species and rank abundance distributions and compared these with distributions of records in the national databases over the same time. We plotted relationships of the local number of records and regional occupancy and calculated multiscale indices of rarity for all fungal species.

Results: There was a remarkable congruence in the patterns found in the UK and Switzerland. Regional assemblages are characterized by many rare species, whereas few are common (fitting the lognormal distribution). However, at local scales, distributions best fitted a power law, suggesting that habitat availability or dispersal processes might play important roles. Fungi with a high number of local records are densely distributed nationally, but unlike other organisms, locally rare fungi may also be densely distributed at a wider scale.

Main conclusions: Fungal fruiting records can be used to infer patterns in fungal distributions. Abundances in local assemblages may be determined by the position of the assemblage in the overall geographical range of each species, dispersal ability and environmental filtering. We advocate the use of multiscale approaches to rarity in future fungal sampling programmes, to provide more reliable information for future conservation policy decisions and fungal biogeography.

KEYWORDS

abundance–occupancy, conservation, fruit bodies, lognormal, models, mushrooms, rank abundance

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1 | INTRODUCTION

Understanding why some species are rarer than others is a fundamental part of community ecology with ramifications in conservation biology, including the management of habitats and natural resources. However, “rarity” itself is a relative concept, in which the abundance or extent of occurrence of a species is defined in relationship to that of others (Gaston, 1994). Furthermore, the rarity of a species depends upon the ecological scale at which the taxon is recorded; species may be considered rare at one scale, but common at another (Hartley & Kunin, 2003). Thus, rare species should never be considered in isolation, but as important components of assemblages and “hotspots” of species diversity (Heegaard, Gjerde, & Saetersdal, 2013).

The species abundance distribution (SAD) is one of the simplest ways of describing the pattern of relative abundance across the species detected in an assemblage. Fisher, Corbet, and Williams (1943) realized that histograms depicting the frequencies of species abundance show a hollow curve, in which many species consist of a few individuals, whereas only a few species are abundant. This pattern appears to be universal and thus, in alternative parlance, most species are rare, whereas only a few are common (McGill et al., 2007). A useful complementary method of describing community assemblages is the rank abundance distribution (RAD), in which a species' abundance is plotted against their rank in abundance (Foster & Dunstan, 2010). Rank abundance distributions can be informative, because they display all the data rather than grouping abundance into “bins”, resulting in the masking of some information (McGill et al., 2007).

A large body of work exists on fitting models to the hollow curve. However, few models are ever rejected, because their prediction ends with stating the nature of the curve, and little attempt has been made to go beyond this and provide explanations for it (McGill et al., 2007). Nevertheless, SADs remain one of the most important tools for describing and understanding community assembly and its management. In particular, they can be useful in informing conservation decisions and determining extinction risk (Matthews & Whittaker, 2015).

It is evident that although SADs have been produced for virtually all groups of organisms, their use in fungal ecology is extremely restricted. In a meta-analysis of SADs, only two (of 558 distributions) were of fungi, both involving lichens (Ulrich, Ollik, & Ugland, 2010). Lichenized fungi, which are macroscopic organisms, are often treated like plants in ecological studies, but non-lichenized fungi are often studied completely separately. This is at least partly because non-lichenized fungi are invisible for most of their lives as mycelium within soil, wood or some other solid substrate, making enumeration almost impossible (Watkinson, Boddy, & Money, 2015). However, ascomycetes and basidiomycetes periodically produce macroscopic fruit bodies, which can be enumerated.

The development of molecular techniques to detect hidden mycelia is revolutionizing the study of fungal communities and has provided indications of patterns in global fungal biogeography (Tedesco et al., 2014). However, SADs and RADs depend on

recording the numbers of individuals of each species, which is still not practicable on a large scale using molecular approaches.

Based on the meta-analysis of macroorganisms other than fungi, Ulrich et al. (2010) concluded that fully censused assemblages tend to show SADs that are best described by a lognormal distribution, whereas assemblages that are incompletely sampled tend to show distributions that are best described by the log series model or a type of power law. These latter distributions still show extreme skew (very many rare species), even when plotted on a logarithmic scale. More recently, Dumbrell, Nelson, Helgason, Dytham, and Fitter (2010a) examined the abundance of arbuscular mycorrhizal fungi (AMF) in 33 different communities, testing three different models: the lognormal fitted in 73% of cases; 27% were described by the broken stick model and none by the geometric series. Unterseher et al. (2011) also found that the lognormal model best described AMF abundance, and Dumbrell, Nelson, Helgason, Dytham, and Fitter (2010b) showed that the lognormal model also described AMF abundance well, but the best model fit was a neutral model, the zero sum multinomial. This led to the conclusion that niche differentiation processes are important in structuring the community, in addition to neutral processes, such as dispersal limitation. These findings contrast with small-scale studies of fungi on leaves, where log series models (indicating incomplete sampling) predominate (Thomas & Shattock, 1986; Unterseher et al., 2011), although neutral models have also provided a good fit (Feinstein & Blackwood, 2012).

Although the majority of fungal studies have taken place at localized scales, those that used broader scales also showed variation in best model fits, with lognormal or log series (Nielsen, Kjoller, Bruun, Schnoor, & Rosendahl, 2016) or neutral models (Gumiere, Durrer, Bohannan, & Andreote, 2016). Thus, it is unclear whether fungi do or do not show similar patterns of abundance to other organisms, beyond the fact that most species seem to be rare and few are common (Nemergut et al., 2013). In particular, given that rarity is scale dependent, a true depiction can be achieved only by multiscale comparisons (Leroy, Canard, & Ysnel, 2013), yet such an approach has never been applied to fungi.

The interrelationship of rarity and scale is formalized in another long-established pattern in macroecology: the abundance–occupancy relationship (i.e., the relationship between the local abundance of species and the size of their ranges within a region). A large body of literature, both theoretical and empirical, has shown that this is another universal and positive relationship; species that are locally abundant tend to occupy wider ranges (i.e., they are more widespread; Borregaard & Rahbek, 2010; Gaston et al., 2000). However, although a number of comprehensive reviews list the wide variety of taxonomic groups that show such relationships (e.g., Holt, Gaston, & He, 2002), fungi are absent from all such analyses. It is intriguing that the closest approach is that of Roney, Kuparinen, and Hutchings (2015), in which lichens were the only taxonomic group in Canada not to show a positive relationship, although sample size ($n = 15$ species) was small.

Abundance–occupancy relationships are not only important in understanding the structuring forces in local communities, but also for their role in species conservation and habitat management (McGeoch & Latombe, 2016). These relationships, in addition to analysis of SADs and RADs, could provide essential tools for characterization of rare fungi and their conservation, but to date, such an approach is lacking. In general, for the very same reasons of cryptic and sampling outlined above, fungi have lagged behind most taxonomic groups in assessments of rarity and the construction of Red Lists for their protection (Dahlberg, Genney, & Heilmann-Clausen, 2010). However, many species of fungi produce fruiting structures (sporocarps) that can be counted, thereby providing a good proxy for individual abundance, and from which designations of “common” or “rare” can be inferred (Dahlberg & Mueller, 2011). Although assigning fruit bodies to different individuals may be problematic at very small scales (< 10 m; Dahlberg & Mueller, 2011), records of occurrence across geographical ranges, calculated as the “area of occupancy” (Gaston, 1994), can provide data that are of similar quality to other taxonomic groups (Truong et al., 2017). The limiting factor is then the distribution and knowledge of recorders, but coordinated large-scale surveys have enormous untapped potential to provide information on fungal species abundance and distributions, which we currently lack (Andrew et al., 2017; Molina, Horton, Trappe, & Marcot, 2011). Indeed, surveys of fruit bodies are an accepted method for biodiversity assessments and often reveal species that are undetected by sequencing methods (Runnel, Tamm, & Lohmus, 2015). Furthermore, while offering great promise for the future, molecular methods cannot currently be used to perform macroecological studies of the types reported here. This is because of the problems that exist within sequence databases, owing to the high number of unnamed and incorrectly named species, plus primer and other methodological biases, which are particularly acute for soil-dwelling species (Khomich et al., 2018).

Coordinated databases of the occurrence of fungal fruit bodies have been used successfully to document recent changes in the phenology and spatial distributions of fungi, in response to changing climate (Boddy et al., 2014; Gange et al., 2018). Similar databases and museum collections have been used in a wide variety of plant and animal studies to examine distributions of species abundances and ranges, to show changes over time and, ultimately, to inform conservation policy (Pyke & Ehrlich, 2010). However, with the exception of a couple of notable animal and plant pathogens, fungi are again absent from such analyses.

Here, we use two databases of fungal fruiting records, which include lichenized fungi, assembled in the UK over the last 65 years: (a) a local data set comprising haphazard collections, with at least weekly frequency from 1950 to 2014, covering an area of c. 3,000 km², part of which was originally used to document phenological changes (Gange, Gange, Sparks, & Boddy, 2007); and (b) data for the whole of the UK, taken from The Fungal Records Database of Britain and Ireland (FRDBI; www.fieldmycology.net). We also use two data sets from Switzerland (which do not include lichenized fungi); a local study in five plots, each of 300 m² (three

10 m × 10 m) plots at the La Chanéaz Forest Reserve, comprising weekly fruit body counts from May to December from 1975 to 2006 (described by Andrew et al., 2018), and data for the whole of Switzerland over the same period (www.swissfungi.ch). A part (28 years) of the local data set was used in a general analysis of varying bin sizes and species abundance distributions (Straatsma & Egli, 2012). To our knowledge, no other local data sets in the world are as comprehensive in their extent and time span as these (Andrew et al., 2017).

Our first objective was to examine the SAD and RAD from each of these data sets, using fungal records to calculate abundance on the basis of both the number of records and the occupancy. Our hypothesis was that national scale data would show the classic log-normal distribution, indicating “complete sampling”, whereas the local sets might show different (log series or power law) patterns, suggesting dispersal limitation or niche-related processes (Ulrich et al., 2010). Our second objective was to examine the abundance–occupancy relationships for these fungi, and we hypothesized that these would be positive, given that other microbial abundance patterns, such as species–area relationships, seem to mirror those of other organisms (Nemergut et al., 2013). Finally, we examined the multiscale patterns of rarity in fungi in the UK and Switzerland, using the approach of Leroy et al. (2013). In general, species that are rare at a local scale also tend to be rare at a wider, regional scale (Freckleton, Gill, Noble, & Watkinson, 2005). If the long-established assertion that “everything is everywhere, the environment selects” (O'Malley, 2008) is correct, then fungi should follow a similar pattern. However, fungi are often dispersal limited (Molina et al., 2011; Peay, Kennedy, & Talbot, 2016), meaning that the relationship might be far less obvious.

2 | METHODS

2.1 | Composition of the data sets

The UK local data set consists of 62,087 occurrence records of 2,319 fungal species, spanning the years 1950–2014 (inclusive). A total of 310 observers contributed records from 1,558 different localities, all within a 30 km radius of Salisbury, Wiltshire, UK (51.068° N, 1.795° W), covering an area of 2,828 km². All records and identifications were confirmed by the late E. G. Gange, with problematic species being confirmed by the Royal Botanic Gardens, Kew. Localities were visited on a haphazard basis, but each was visited at least once per year. Each record was referenced by a six-figure Ordnance Survey grid reference (<https://www.ordnancesurvey.co.uk/>), allowing occupancy to be calculated at a resolution of 1 km × 1 km squares (out of a total of 614). Further details on record assembly are provided by Gange et al. (2007). “Abundance” was defined as the total number of records of occurrence (i.e., not the total number of fruit bodies) for each species, and area of occupancy as the total number of 1 km × 1 km squares in which each species was recorded, so as to be comparable with similar previous analyses (Gaston et al., 2000).

The full UK national data set spans a wider time-scale and also contains records for the island of Ireland. These data are from multiple sources, contributed by individuals, foray lists, scientific societies, herbaria records, and publications of the British Mycological Society. We excluded records collected: (a) before 1950; (b) from outside mainland Britain; and (c) that were missing location data, leaving 1,361,069 separate data points for the 2,319 species from 55,882 localities over an area of c. 209,330 km². These records do not include the local data (above), which have since been incorporated into the national set. Each record was referenced at the resolution of 10 km × 10 km; the standard grid system in the UK. As above, “abundance” was measured as the total number of records for each species, whereas occupancy was the number of 10 km × 10 km squares in which each species was found. A complete list of all species used is given in Dryad: <https://doi.org/10.5061/dryad.h36dj4t>.

Sampling for the Swiss local data set is described in full by Heegaard et al. (2017) and Andrew et al. (2018), and full species lists are given in the supplementary material of Andrew et al. (2018). Weekly fruit body counts took place from 1975 to 2006, but the 10 m × 10 m plots were divided into 1 m × 1 m contiguous sub-plots only in 1992. Thus, the description of occupancy is not relevant in this data set, because the scale is too small and the data consist of the total number of records of 319 separate species. A subset of the complete Swiss national data set was used, to include the same 319 species over the same time span as the local data, producing 97,358 separate records covering an area of 41,285 km². Occupancy at the 10 km × 10 km scale was defined in the same way, using grid references as above. A list of all data sources is given in the Supporting Information (Appendix S1).

2.2 | Data analyses

2.2.1 | Local data set species accumulation curves

We first examined the nature of the local data sets by calculating species accumulation curves and estimating the species pool that might exist if all possible species were found. The latter was examined with three methods: Chao 2, first-order jackknife and bootstrap. These analyses were performed with the package “vegan” in R v.3.4.1 (Oksanen et al., 2017; R Core Team, 2017).

2.2.2 | Species abundance distributions

All species abundance distributions were fitted using the R package “sads” (Prado, Miranda, & Chalom, 2016), which uses maximum likelihood methods to fit and compare different models. These were the gamma, lognormal and Weibull (the three most commonly used continuous distributions), plus the geometric and the negative binomial models. These were fitted with zero truncation, because species with zero records were unknown. We also fitted Fisher's log series and three associated power law functions, Pareto, power and power bend (implementation of the latter two not including zeroes); the log series being a special case of the power bend. We also fitted MacArthur's broken stick model and the Poisson lognormal model;

the latter describes species abundances in a Poisson sample from an underlying lognormal assemblage. Finally, we examined the fit of two neutral models, the metacommunity zero-sum multinomial distribution, which is thought to describe a community under random drift, and the Volkov model, which is thought to describe a community under neutral drift, with immigration. References to the use of all models are provided by Prado et al. (2016).

2.2.3 | Rank abundance distributions

Rank abundance distributions were fitted using the “radfit” command in the R package “vegan” in a similar manner. In this case, we used the niche pre-emption model (also termed the geometric series), lognormal, broken stick, and two discrete power law distributions: the Zipf and Zipf–Mandelbrot. In all cases, Akaike information criterion (AIC) values were computed to determine which model(s) provided the best fit to the data (Burnham & Anderson, 2002). Those produced for each model were compared using the delta AIC (Δ_i). A Δ_i value < 2 indicates interaction between models, values $3 < \Delta_i < 9$ indicate little interaction, and $\Delta_i > 10$ no interaction (Burnham & Anderson, 2004).

2.2.4 | Relationships between abundance at local and national scales

Relationships between the number of records at local and national scales were examined with a generalized additive model (GAM) procedure, using the “mgcv” package in R. To examine abundance–occupancy relationships, we followed Holt and Gaston (2003) and Zuckerberg, Porter, and Corwin (2009) in using the logit transformation for occupancy data and the $\log_{10}$ -transformation for the number of records, expressed as the total number for each species over the 65 and 32 year periods. As Zuckerberg et al. (2009) comment, phylogenetic approaches are rarely required in abundance occupancy analyses, because closely related species can vary greatly in their distributions and population sizes. As a check, we examined some of the larger genera represented in the UK databases, *Cortinarius* (88 species), *Russula* (72 species), *Mycena* (59 species), *Lactarius* (44 species) and *Entoloma* (44 species). For these genera, the record data at the local scale showed huge variation, with a ratio between the most and least numerous species of 34, 100, 104, 64 and 36 times, respectively. Thus, even at the local scale, the variation in the number of records of species within genera is so great that controlling for phylogenetic relatedness is unwarranted, and we conducted an ordinary least squares regression procedure, with occupancy as the dependent variable (Holt & Gaston, 2003).

2.2.5 | Multiscale rarity

If nonlinear relationships between abundance at different scales are found, then a multiscale approach is justified (Leroy et al., 2013). As this was so, we calculated two-scale (local and national) rarity weights for all species in the UK and Switzerland, using the R

package “rarity” (Leroy et al., 2013). These were scaled between zero and one, with a greater degree of rarity being indicated by a higher value of the index. Calculation of these weights involves an appropriate weighting method and the use of a rarity cut-off value. To achieve this, we used Gaston's quartile definition, that rare species are those 25% with the lowest number of records or occurrence (Gaston, 1994; Leroy et al., 2013).

3 | RESULTS

3.1 | Characteristics of the local data sets

There was no trend over time in the number of collections or forays per year in the UK (Figure 1a; $F_{1,63} = 3.51$, $p > 0.05$). It should be noted that databases of this type do not record forays when no species were found. Low numbers in years such as 1976, 1990 and 2011 represent poor years for fruiting, caused by lack of rainfall.

The total number of species (the “species pool”) estimated from the UK species accumulation curve (Figure 1b) varied depending on the method, with the Chao 2 estimating $3,526 \pm 96.9$, jackknife $3,313 \pm 141.4$ and bootstrap $2,852 \pm 80.8$. Therefore, these values suggested that between 68 and 84% of the total “available” species were detected in the 65 years of recording.

The total species pool estimates for the Swiss local data were more consistent, with the Chao 2 estimating 388.75 ± 21.8 , jackknife

388.75 ± 17.4 and bootstrap 351.19 ± 10.7 . These values suggested that 82–90% of the total species were recorded in the 32 years of sampling. The species accumulation curve is shown in Figure 1c.

3.2 | Species and rank abundance distributions

The species abundance distribution for UK national scale records was best fitted by the lognormal distribution (AIC = 33,305.1; Figure 2a). No other models provided a good fit to the data (Supporting Information Table S1), with the next best fit provided by the Poisson lognormal ($\Delta_i = 18.0$, $p < 0.001$). There was no interaction between models ($18 < \Delta_i < 3,592$). The SAD for national scale occupancy data was also fitted best by the lognormal model (AIC = 29,093.6; Figure 2b), this being better than the Weibull model ($\Delta_i = 9.2$, $p < 0.01$). The rank abundance distribution for national record number was also best fitted by the lognormal model (AIC = 84,952; Figure 2c), but that for occupancy data was best fitted by the pre-emption model (geometric series; AIC = 24,202; Figure 2d). With both RADs, fitting of the Zipf–Mandelbrot failed, because its estimation is difficult (Oksanen et al., 2017).

The Swiss national record number data were also best fitted by the lognormal distribution (AIC = 3,997.6), although the Weibull and broken stick also provided a good fit (Supporting Information Table S1; Figure S1). The SAD for Swiss national scale occupancy data was also best fitted by the lognormal model (AIC = 3,248.4). The rank

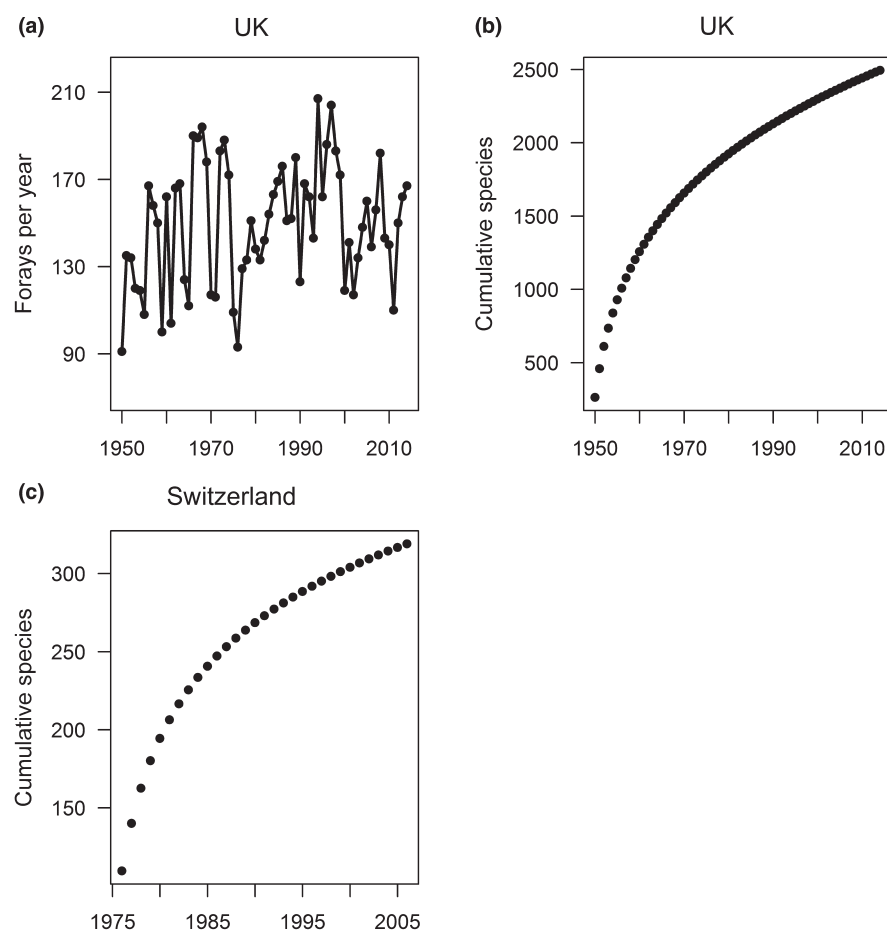


FIGURE 1 Temporal characteristics of the local data sets. (a) The number of annual forays in the UK. (b) The accumulated total number of species observed over 65 years in the UK. (c) The accumulated total number of species observed over 32 years in Switzerland

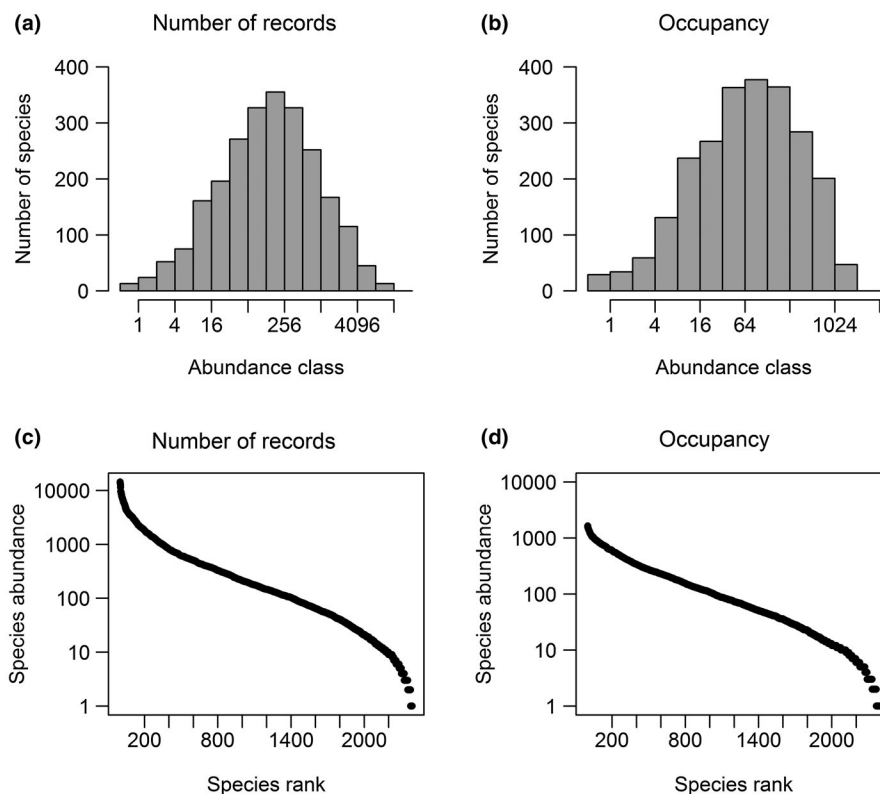


FIGURE 2 Frequency distributions for species abundance (SAD) and rank abundance (RAD) of the UK national data set based on: (a,c) number of records; and (b,d) occupancy

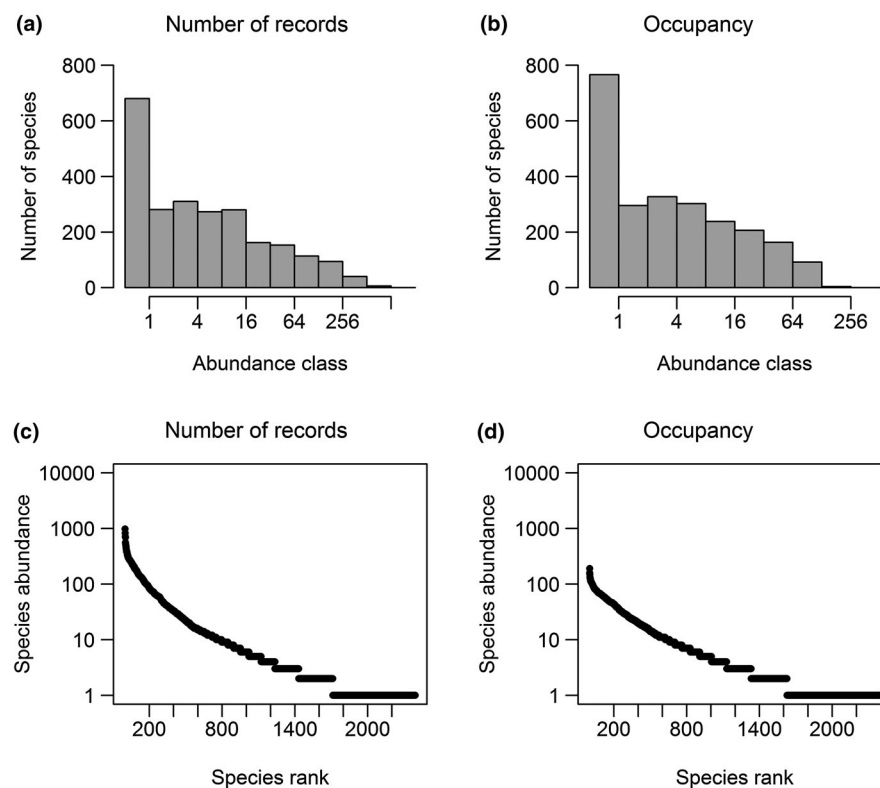


FIGURE 3 Frequency distributions for species abundance (SAD) and rank abundance (RAD) of the UK local data set based on: (a,c) number of records; and (b,d) occupancy

abundance distribution for Swiss national record data was best fitted by the broken stick, whereas occupancy data were best fitted by the geometric series (Supporting Information Table S1).

The species abundance distribution for UK local record numbers was best fitted by the Pareto distribution (AIC = 15,594.8;

Figure 3a), as was that for the Swiss local record numbers (Supporting Information Table S1; Figure S1). Meanwhile, the abundance distribution for UK local occupancy was also best fitted by the Pareto distribution (AIC = 13,367.4; Figure 3b; Supporting Information Table S1). In both cases, all other models differed from the Pareto

($p < 0.001$). Rank abundance fits for UK local data followed an identical pattern to national data (Supporting Information Table S1), fitting a lognormal and geometric series, respectively (Figure 3c,d). Swiss local record data differed from the national data, wherein the lognormal provided the best fit (Supporting Information Table S1).

3.3 | Abundance–occupancy relationships

Species with the greatest number of records at the UK national scale also had the greatest number at the local scale (Figure 4a). These data were best fitted by a nonlinear model ($R^2 = 58.8\%$, $p < 0.001$), rather than a linear relationship ($R^2 = 47\%$), because species with very few records (i.e., were “rare”) locally might have few or very many records nationally, illustrated by the flat bottom to the graph. However, there appeared to be a tipping point, with very common species ($> 1,000$ records nationally) showing a linear relationship with the local number of records. A similar relationship was seen for occupancy data; species that are most densely distributed at the national scale are also so at the local scale (Figure 4b), but locally sparsely distributed species might be sparsely or densely distributed nationally. A nonlinear

relationship ($R^2 = 57\%$, $p < 0.001$) was also seen in these data, rather than a linear one ($R^2 = 44.4\%$), with a tipping point of occurrence in c. 200 of the national $10 \text{ km} \times 10 \text{ km}$ squares.

Swiss record data followed a very similar pattern (Supporting Information Figure S2) and were best fitted by a nonlinear model ($R^2 = 21.1\%$, $p < 0.001$), rather than a linear relationship ($R^2 = 15.1\%$). Again, there appeared to be a tipping point, with very common species (> 350 records nationally) showing a linear relationship with local record number (Supporting Information Figure S2).

The abundance–occupancy relationships showed a remarkable similarity in the two countries (Figure 5). Although both relationships are significant, they are relatively weak (UK: $R^2 = 43.9\%$; Swiss: $R^2 = 13.8\%$), because species that are rare (least abundant) on a local scale can be sparsely or densely distributed on a national scale. In contrast, species with high local record numbers tend to be densely distributed nationally. Species with high local record numbers but sparse distributions nationally are absent in both data sets.

3.4 | Multiscale indices of rarity

In the UK, most, but not all, species that are rare on a national scale are also rare on a local scale ($R^2 = 65.2\%$, $p < 0.001$; Figure 6a), and the significant relationship is clearly driven by the preponderance of data with high rarity indices. Likewise, those which are sparsely

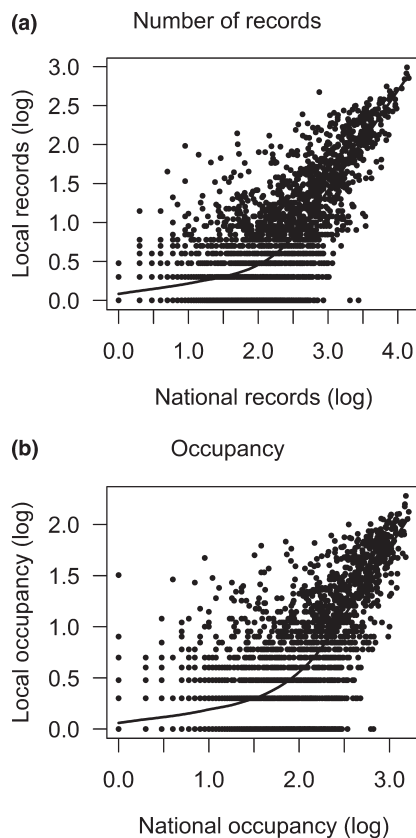


FIGURE 4 Relationships between UK local and national scale data. (a) Relationship between number of local and national records. (b) Relationship between local and national occupancy. Species with the most records nationally also have the most records at a local scale (a), whereas those that are most densely distributed nationally are also so at a local scale, measured by counts in 10 and 1 km grids, respectively (b). The fitted lines are from a locally weighted regression (LOESS) procedure

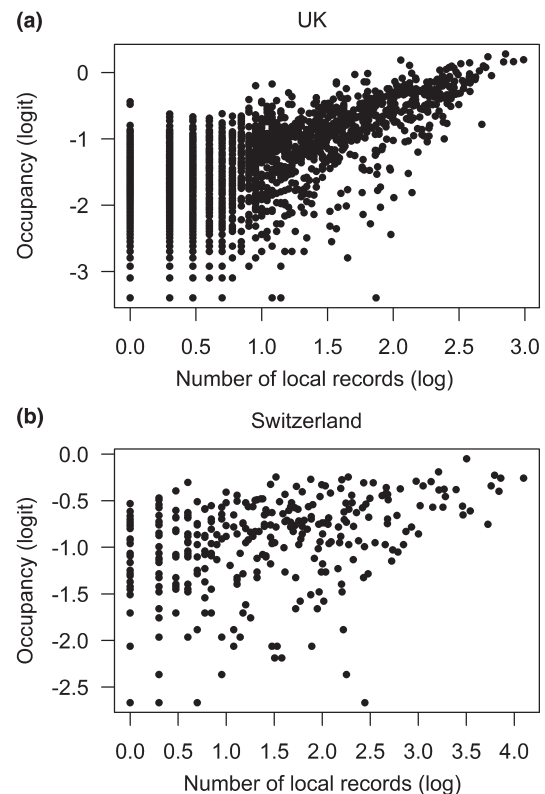


FIGURE 5 The abundance–occupancy relationships for: (a) British fungi; and (b) Swiss fungi. Species with high numbers of records locally tend to be distributed densely at a wider scale. However, species with low local record number may be distributed sparsely or densely at a wider scale

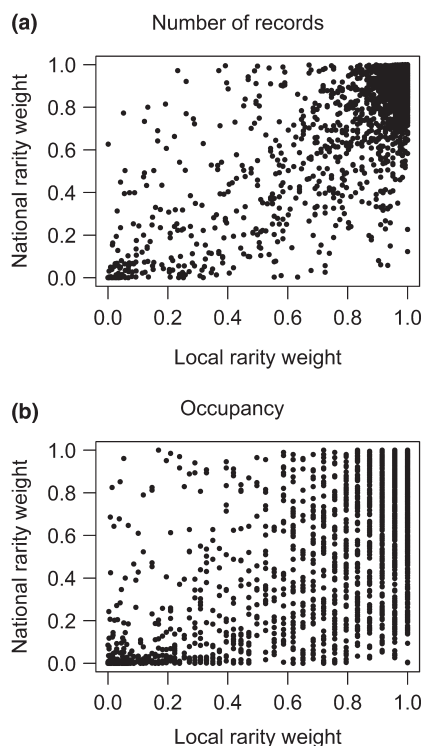


FIGURE 6 Multiscale patterns of rarity in British fungi. Using number of records (a) or occupancy (b), species can be rare at one scale but common at another, and vice versa. The majority of species are rare at both scales, using data of the number of records, but the pattern for occupancy is different, suggesting that species with low occupancy locally might be distributed sparsely or densely at a larger scale

distributed nationally also tend to be sparsely distributed on a local scale (Figure 6b), although the relationship is considerably weaker ($R^2 = 49.8\%$, $p < 0.001$) and the pattern more diffuse. The pattern is even more accentuated in Swiss record data (Supporting Information Figure S3; $R^2 = 12.3\%$, $p < 0.001$), with the majority of species being rare at the local scale. In these data, there were no species that were locally common but nationally rare.

The proportions of species falling into the four possible categories of rarity generated by the two-scale approach are given in Table 1. For both data sets, the vast majority of species can be considered rare at both spatial scales. Only between 1% (based on the number of records in the UK and Switzerland) and 6% (based on occupancy in the UK) of species could be considered common at both spatial scales (Table 1).

4 | DISCUSSION

This is the first macroecological study of rarity patterns in fungi. By using four comprehensive data sets, we have shown that although some similarities exist in patterns of abundance between fungi and other organisms, there are also noticeable differences. Both UK and Swiss data showed the classic hollow curve of species abundance,

TABLE 1 Percentage of species in different categories of rarity, as measured by the number of records and occupancy (the latter in parenthesis for UK data only)

		National	
		Rare	Not rare
UK, n = 2,319			
Local	Rare	96.2 (85.3)	1.6 (7.5)
	Not rare	0.6 (1.1)	1.6 (6.1)
Switzerland, n = 319			
Local	Rare	87.3	10.9
	Not rare	0	1.8

but although the national data were best fitted by a lognormal model, the local data sets were uniquely fitted by a Pareto distribution (with very many rare species). Abundance–occupancy relationships of fungi were positive, and similar in the UK and Switzerland, but showed a different pattern to those of other organisms.

Perhaps the most obvious conclusion from these data is that irrespective of the approach taken, most species of fungi are rare, whereas only a few can be considered common. Both the number of records and occupancy at the national scales were best described by the lognormal distribution, as with marine and soil bacteria (Ferrenberg et al., 2013; Fuhrman, 2009; Nemergut et al., 2013) and most macroorganisms (Ulrich et al., 2010). There has been much debate on whether lognormal SADs are the product of sampling artefacts, model fitting, the influence of environmental variables, or the apportionment of niches between species (McGill et al., 2007; Williamson & Gaston, 2005). Furthermore, we must not forget that the lognormal is a purely statistical distribution. However, despite the potential for artefacts, the finding that lognormal distributions tend to arise in assemblages that are completely sampled (Ulrich et al., 2010) suggests strongly that there are good biological reasons for such patterns, including niche processes, competition and dispersal.

The dictum for microbes that “everything is everywhere, but the environment selects” (reviewed by O'Malley, 2008) has been challenged in many microbial studies (Martiny et al., 2006). Fungal species certainly exhibit biogeographical patterns at continental or smaller scales (Taylor, Turner, Townsend, Dettman, & Jacobson, 2006; Tedersoo, 2017). In general, everything is not everywhere, but environmental filtering certainly plays a role in determining fungal distributions, with dispersal limitation being proposed as one of the main drivers (Peay et al., 2016). We found variation in the occupancy within the geographical range of fungi of two or three magnitudes at both national scales and at the local scale in both countries, clearly supporting the fact that not all species occur everywhere within their range. Furthermore, relatively low amounts of variation were explained by our analyses of local versus national numbers of records and occupancy, and also with the multiscale analysis of rarity indices, suggesting that environmental filtering, most probably manifest as habitat availability, plays an important role. This was particularly true in Switzerland, where landscape structure is more heterogeneous than in the UK (Hofer, Wagner, Herzog, & Edwards, 2008).

A recent meta-analysis suggests that few fungi may be habitat generalists (Meiser, Bálint, & Schmitt, 2014) and therefore the availability of habitats in the local environment determines species occurrence (Kivlin, Winston, Goulden, & Treseder, 2014). This fact is further supported by the two local data sets displaying a fit to a power law, rather than the lognormal, upholding our first hypothesis and suggestive of niche-related processes (i.e., habitat availability) playing a role (Ulrich et al., 2010). Both these data sets are free from many forms of bias that influence such analyses (Gange, Gange, Mohammad, & Boddy, 2012; Lavoie, 2013), particularly given that they were coordinated regular surveys over long time periods, which were not influenced by citizen scientists recording certain species (e.g., edible fungi) or searching in known localities (Geldmann et al., 2016). However, such model fits might still be indicative of incomplete or imperfect sampling (McGill, 2003; Ulrich et al., 2010), particularly given that neither species accumulation curve was asymptotic. A further complication is that spatial distributions of saprotrophic and ectomycorrhizal fungi in the UK have changed in recent years, correlated with changes in temperature and rainfall (Gange et al., 2018).

This is the first time that a Pareto distribution has been fitted to fungal data sets, although previous authors generally examined a limited range of models (e.g., Dumbrell et al., 2010a; Unterseher et al., 2011; Gumiere et al., 2016). Pareto distributions are characterized by the presence of many species of low abundance (often singletons) and far fewer of high abundance, producing a long, "thin" tail to the SAD. Such distributions are widely reported in physics, computer science, economics and social sciences in addition to biology (Newman, 2005). Our data are remarkably similar to those of many invertebrate distributions, with 28% singletons in the UK local fungal data, and 9% in the Swiss data, compared with 29–32% for tropical arthropods (Coddington, Agnarsson, Miller, Kuntner, & Hormiga, 2009) and 17% for temperate spiders (Leroy et al., 2013). However, this is considerably less than the 53% singletons recorded in a short-term (1 year) survey of fruit body abundance of temperate forest fungi in northern Spain (Abrego & Salcedo, 2014).

It is most likely that dispersal ability and niche availability influence the numbers of very rare fungal species (Unterseher et al., 2011). In British lichens, species with high colonization ability (dispersal) occupy larger geographical ranges (Leger & Forister, 2009), and this might be true for fungi generally (Cox, Newsham, Bol, Dungait, & Robinson, 2016); therefore, limited dispersal ability might explain why many species are rare (Molina et al., 2011). Meanwhile, given that the number of fungi in an assemblage might be functionally related to the number of plants (Hawksworth, 1991; although for a contradiction with soil fungi, see also Tedersoo et al., 2014), and assemblages of plants, even when completely censused, appear to follow power law distributions (Ulrich et al., 2010), the availability of plant substrates (niches) might be a major factor influencing fungal distributions.

Fungal species that were rare (measured as either records or occupancy) on a national scale were also rare on a local scale. The national number of records explained c. 47–58% of the variation in local numbers of records, similar to the 31% for phytoplankton

and 58% for aquatic bacteria (Ostman et al., 2010). This conclusion was also reinforced by the multiscale analyses; here, the majority of species were rare at both scales. However, this analysis showed that it is possible for all parts of the graphical spectrum to be occupied, further refuting the "everything is everywhere..." idea and upholding our original hypothesis. Calculation of the rarity weights for each species is an important aspect of our analysis, because it gives a numerical index for each species, which is far more informative than simple categories (Leroy et al., 2013). In future, such indices could help with the construction of Red Lists, provided that the quantitative information can be related to International Union for Conservation of Nature criteria adapted for fungi (Dahlberg & Mueller, 2011).

Our study is also the first to find good evidence for a positive abundance–occupancy relationship for fungi, upholding our second hypothesis. Comparison with previous studies is hampered by authors using different measures of occupancy and abundance or by treating abundance as the dependent variable. However, the form of the fungal relationship is different to that of other analyses which have used the same approach (logit and log, with occupancy on the y axis; e.g., Holt & Gaston, 2003; Zuckerberg et al., 2009), which show clear linear relationships, with R^2 values between 60 and 90%. The fungal relationships, although very similar in the two countries, were much more diffuse and bear some similarity to that of marine bacteria (Amend et al., 2013), but not to intestinal bacteria (Green, Fisher, McLellan, Sogin, & Shanks, 2016). Our data are similar to all others in that there were no locally abundant species with sparse national distributions, but the crucial difference is that there were many locally rare species that were densely distributed at a wider scale.

Eight possible mechanisms that might account for a positive abundance–occupancy relationship have been proposed (Gaston et al., 2000; Gaston, Blackburn, & Lawton, 1997). The first pertains to sampling bias, which may result from low sampling intensity at small spatial scales. We do not believe that poor sampling in the local data sets has contributed to the observed relationship, because there was no trend in "foray effort" over time in the UK, and standardized weekly counts were conducted in Switzerland. Furthermore, the UK local data comprised 6,868 separate forays, and each involved the collection and identification of every fungus seen, whereas the Swiss data comprised 992 separate sampling occasions. Citizen science data, although not without its faults, can be used for macroecological analyses, if collated properly, and many previous analyses of this type have used such data (Dickinson, Zuckerberg & Bontner, 2010). Indeed, patterns of phenology in these national and local data sets are remarkably similar (Andrew et al., 2018), suggesting that the local data sets were not biased towards or against certain species. It is possible that model fits might change if all the local species estimated to be in the pool were found. However, given that all the "missing" species must be represented by very low numbers of records, the most likely outcome would be that the model fit would remain the same, while the thin tail of the SAD would be extended.

Phylogenetic relatedness has been suggested as a second possible, but unlikely, influence on the relationship (Gaston et al., 1997)

and is also unlikely in our study owing to the enormous variation in abundance and range within almost all genera. A third proposal is that the position in the overall geographical range of a species is important in determining its local abundance; species at the edge of their overall range generally have lower abundance. It is interesting that a prediction arising from this hypothesis is that widespread species may show high or low local density, whereas species with restricted geographical ranges have only low density (Gaston et al., 1997). This would give a triangular abundance–occupancy relationship, as found in the present study. Given the differences in UK climate from north to south and from west to east, and the sensitivity of fungi to climate (Boddy et al., 2014; Gange et al., 2018), it is plausible that locally rare species might vary greatly in their geographical range, depending on what part of the overall range the local area occupies. The other five hypotheses, based on resource use, resource availability, habitat selection, metapopulation dynamics or vital rates, seem far less likely to apply to fungi.

Overall, using four extensive data sets, we have shown that fungi exhibit some markedly different macroecological patterns to other organisms. In particular, the abundance–occupancy relationships for fungi are very different and suggest that the forces that determine commonness and rarity in other organisms are different for fungi. It is clear that after either 32 or 65 years of intensive sampling in the two localities, many fungal species remained undetected and that the vast majority of species could be considered as rare.

ACKNOWLEDGMENTS

We are grateful to the many recorders who contributed to the national data sets and to the late Mrs I. Y. Gange and the 309 other observers who contributed to the UK local data set. Mr François Ayer and staff at the Swiss Federal Research Institute (WSL) collected the Swiss local data set.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

AUTHOR CONTRIBUTIONS

A.C.G. and E.G.G. designed the study. E.G.G. collected the local data in the UK, which were analysed by A.C.G., L.P.A. and A.N. S.E. collected and collated the local data in Switzerland. B.S.-I. coordinated and georeferenced the Swiss national data. C.A. harmonized, formatted and prepared all the Swiss data, which were analysed by A.C.G. A.C.G., L.P.A., C.A. and L.B. wrote the paper, with contributions from all other authors.

DATA ACCESSIBILITY

All data used in this study are available in the Dryad depository, <https://doi.org/10.5061/dryad.h36dj4t>

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BIOSKETCH

ALAN C. GANGE and colleagues are interested in the use of historical databases to examine spatial and temporal trends in fungal fruiting. The UK local data set was the life's work of Alan's late father, who began recording fungi in 1950 because at that time no distribution maps existed. These data have revealed dramatic changes in fungal phenology and range shifts over the last 65 years.

SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

How to cite this article: Gange AC, Allen LP, Nussbaumer A, et al. Multiscale patterns of rarity in fungi, inferred from fruiting records. *Global Ecol Biogeogr*. 2019;28:1106–1117. <https://doi.org/10.1111/geb.12918>